

新莊五大水系防災導引 GIS 平台

Flood Evacuation Guidance System for the Five Tributaries in Xinzhuang District

System Plan

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System Specification

Requirement Definition

Flood Evacuation Guidance System for the Five Tributaries in Xinzhuang District is a web-based disaster evacuation guidance system built on **Geographic Information System (GIS)** technology. It is designed to provide evacuation guidance and disaster-related information services for flooding scenarios caused by heavy rainfall in **Xinzhuang District, New Taipei City**.

Xinzhuang District contains five major tributaries—**Zhonggang Drainage, Taliaokeng Creek, Tandidi Creek, Shibafenkeng Creek, and Yakoukeng Creek**—which are frequently flooding areas during rainy seasons. Conventional navigation applications prioritize the shortest distance or fastest travel time when planning routes, without considering riverbank or flood-prone areas. To address this limitation, the proposed platform integrates GIS spatial analysis, real-time rainfall monitoring, and a customized A* disaster-avoidance routing algorithm to generate evacuation routes that prioritize safety over travel efficiency.

The system combines data from the OpenStreetMap road network, real-time rainfall observations from the Central Weather Administration (CWA), official evacuation shelter information provided by the New Taipei City Government, and flood potential maps. By integrating these datasets, the platform helps users quickly identify safe evacuation routes and suitable shelters during flood emergencies.

Development Environment

- Target web browsers:

- Google Chrome
- Microsoft Edge
- Apple Safari
- Development tools
 - HTML5
 - CSS3
 - JavaScript
 - Vite
- GIS and Mapping Technologies:
 - Google Maps JavaScript API
 - Google Places API
 - Geocoding API
 - OpenStreetMap
 - GeoJSON
- External Data Services
 - 中央氣象署開放資料 Central Weather Administration Open Data API
 - OpenStreetMap Overpass API
 - 新北市政府開放資料平台 New Taipei City Government Open Data Platform
- Coordinate System
 - WGS84 Geographic Coordinate System
- Data Standards

The spatial data used in this system are stored and exchanged in the GeoJSON format, and all geographic data are represented using the WGS84 coordinate reference system.

The primary data sources of the system are as follows:

1. Spatial data of the five major tributaries and their boundaries
2. Evacuation shelter data for Xinzhuang District
3. Flood potential maps
4. OpenStreetMap (OSM) road network data
5. Real-time rainfall data provided by the Central Weather Administration (CWA)

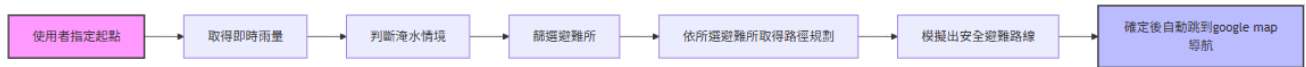
System Scope

Flood Evacuation Guidance System for the Five Tributaries in Xinzhuang District Platform consists of the following three subsystems:

GIS Map Visualization System

Real-time Rainfall Monitoring System

Intelligent Evacuation Guidance System



1. GIS Map Visualization System

This subsystem is responsible for displaying various spatial data layers and providing map interaction functions.

Main layers include:

- Five Tributaries River Layer
- Watershed Boundary Layer
- River Floodplain Layer
- Flood Potential Layer
- Official Evacuation Shelter Layer
- Safe Evacuation Route Layer

2. Real-time Rainfall Monitoring System

This subsystem is responsible for retrieving real-time rainfall information from the Central Weather Administration (CWA) Open Data Platform.

- Main functions include:
- Real-time Rainfall Query
- Rainfall Status Display
- Automatic Data Updates
- Flood Scenario Determination

3. Intelligent Evacuation Guidance System

This subsystem serves as the core component of the project.

Main functions include:

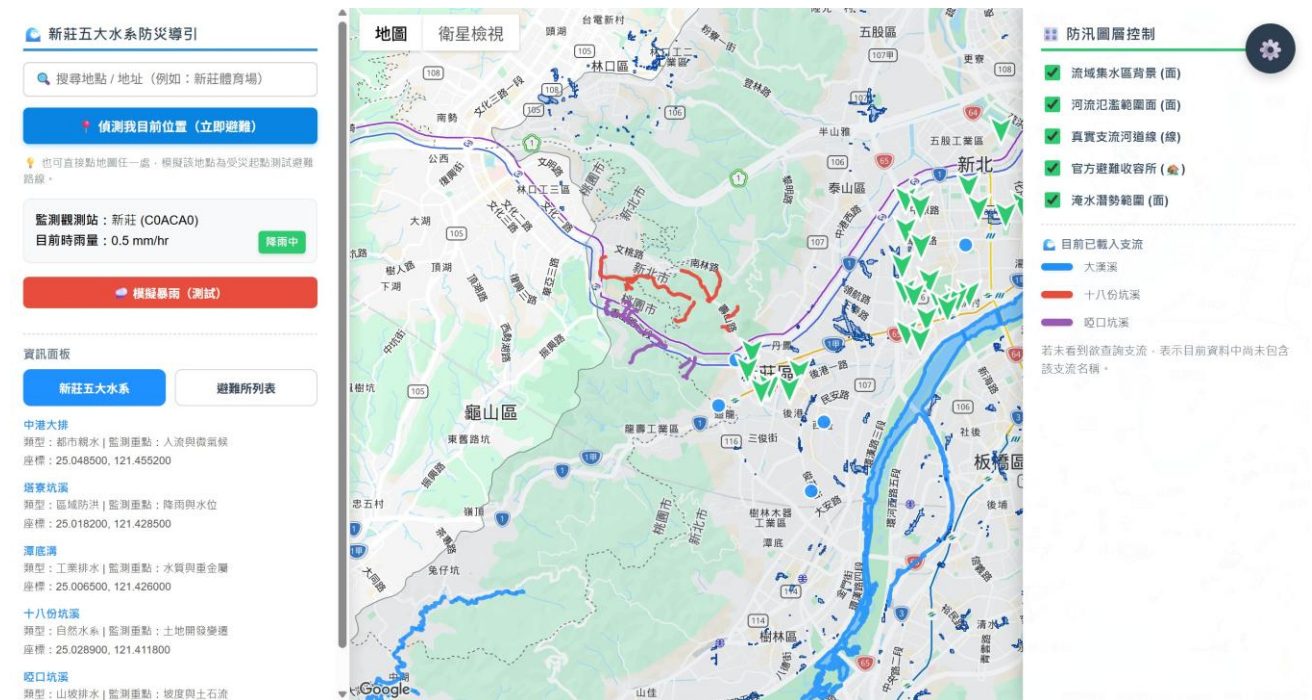
- User Location Detection
- Evacuation Shelter Selection
- Road Network Route Planning
- Disaster-Avoidance Algorithm Processing
- Turn-by-Turn Navigation
- Google Maps Navigation Backup

System Development Plan

1. GIS Map Visualization System

The GIS Map Visualization System provides users with map browsing capabilities and spatial data visualization functions.

Conceptual UI Elements





按下地圖任一位置皆能啟動以當地為出發點的避難策略

Main Functions

1.1 Map View (M0)

The main map interface provides users with access to map browsing, layer visualization, and evacuation guidance functions.

M0-01 地點搜尋框	● M0-01 地點搜尋可輸入地址
M0-02 偵測位置按鈕	● M0-02 GPS 定位按鈕偵測目前位置
M0-03 按下地圖任一位置	● M0-03 按下地圖任一位置皆能啟動以當地為出發點的避難策略
M0-04 “還原真實雨量”/“暴雨模式(測試)”鈕	● M0-04 即時顯示目前雨量，或模擬當地暴雨情況
M0-05 “還原真實雨量”	● M0-05 即時顯示目前雨量
M0-06 模擬暴雨按鈕	● M0-06 模擬當下正下暴雨
M0-07 資訊面板區塊	● M0-07 “新莊五大水”系按鈕 和 “避難所列表”按鈕
M0-08 “新莊五大水系”按鈕	● M0-08 五大水系詳情
M0-09 “避難所列表”按鈕	● M0-09 依據當前位置推薦的避難所列表
M0-10 圖層控制按鈕	● M0-10 圖層控制面板
M0-11 地圖顯示區	● M0-11 所在區域地圖

1.2 Layer Management (M1)

Provides layer visibility control functions.



● Available layers:

流域集水區	• 五大支流河道圖層流域集水區圖層
真實支流河道	• 五大支流河道線圖層
河川氾濫範圍	• 河川氾濫範圍圖層
淹水潛勢區	• 淹水潛勢圖層
官方避難收容所	• 避難收容所圖層

2. Real-time Rainfall Monitoring System

The system obtains real-time rainfall information from the Xinzhuang Rainfall Station via the Central Weather Administration (CWA) Open Data API.

Conceptual UI Elements



按下“還原真實雨量”鈕可以看到即時降雨資訊

按下“暴雨模式(測試)”按鈕可模擬當前暴雨

Main Functions

2.1 Rainfall Monitoring (R0)

Displays current rainfall information.

R0-01 目前觀測站	• 顯示觀測站名稱
R0-02 目前雨量	• 顯示當前雨量(mm/hr.)
R0-03 模擬雨量	• 顯示模擬暴雨情境
R0-04 降雨狀態	• 以醒目顏色顯示目前降雨狀態

2.2 Flood Scenario Selection (R1)

Automatically switches flood scenarios based on real-time rainfall data.

When rainfall data are updated, the system recalculates the evacuation route accordingly.

3. Intelligent Evacuation Guidance System

The Intelligent Evacuation Guidance System integrates road network data, flood potential maps, and evacuation shelter information to provide safety-oriented evacuation route planning services.

Conceptual UI Elements



Evacuation Shelter Selection and Navigation Guidance



Navigation Guidance and Google Maps Backup Navigation.

Main Functions

3.1 Start Location Selection (S0)

Users can specify a starting location using one of the following methods:

GPS Location Detection

Place Search

Map Click Selection

3.2 Shelter Selection (S1)

The system filters evacuation shelters based on the following criteria:

- Located within Xinzhuang District
- Designated for flood evacuation

The nearest qualified evacuation shelter is then selected as the destination.

3.3 Safe Route Planning (S2)

This function serves as the core component of the system.

The system utilizes the OpenStreetMap road network and the A pathfinding algorithm to calculate safe evacuation routes

The hazard factors considered in the route planning process include:

- Distance from riverbanks
- Flood-prone areas
- Real-time rainfall intensity

Cost Function

$$\text{Edge Cost} = \text{Road Length} \times (1 + \text{Hazard Weight})$$

The Hazard Weight is calculated based on the following factors:

- Riverbank Distance Weight
- Flood Area Weight
- Rainfall Factor

These factors are combined to determine the overall hazard weight.

3.3 Step Navigation (S3)

Step-by-step navigation instructions are generated based on the calculated route.

The navigation information includes:

- Forward Direction
- Left-turn Instructions
- Right-turn Instructions
- Remaining Distance
- Estimated Arrival Time

3.4 Google Navigation Backup (S4)



If users require external navigation services, they can directly launch Google Maps navigation as a backup option.

Expected Benefits

Provide safety-priority evacuation guidance services.

- Reduce the risk of the public entering riverbank or flood-prone areas by mistake.
- Integrate real-time rainfall and spatial information to improve disaster response efficiency.
- Demonstrate the integration of GIS spatial analysis and algorithms.
- Serve as a platform for disaster prevention education and presentation purposes.